1 Team Problems

Problem 1. Compute all values of a for which the intersection of the graphs of $y \geq \left\lfloor \frac{x}{2} \right\rfloor$ and $y \leq a|x| + 17$ is a region having area 51.

Problem 2. Let $f(x) = \log_b x$ and let $g(x) = x^2 - 4x + 4$. Given that $f(g(x)) = g(f(x)) = 0$ has exactly one solution and that $b > 1$, compute b.

Problem 3. Compute all solutions to the following system of equations:

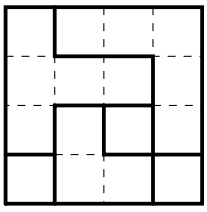
$$\begin{aligned} |y| - \frac{2x}{|x|} &= -1 \\ x|x| + y|y| &= 24. \end{aligned}$$

Problem 4. Two players are playing a game as follows. Integers N and M are each selected independently at random from the set $\{1, 2, \ldots, 50\}$. The players begin with an $N \times M$ grid of empty squares. A turn consists of filling in a single empty square, then filling in all the squares in the same row, and then filling in all the squares in the same column. The players alternate taking turns until all squares are filled in. Compute the probability that the player who goes first fills in the last square.

Problem 5. A sequence $T_1, T_2, \ldots$ is called (a, b)-*nacci* if $T_1 = a$, $T_2 = b$, and for $n \geq 3$, $T_n = T_{n-1} + T_{n-2}$. Compute the number of (a, b)-nacci sequences of positive integers such that $b > a$ and there exists an integer $k \geq 3$ with $T_k = 50$.

Problem 6. Complete the number puzzle below. Clues are given for the four rows. (Answers may not begin with a zero.) Cells inside a region must all contain the same digit, and each region contains a different digit.

1. Product of two primes
2. Multiple of 19
3. A perfect square
4. Multiple of 1-Across



Problem 7. Real numbers x, y, and z are chosen at random from the unit interval $[0, 1]$. Compute the probability that $\max\{x, y, z\} - \min\{x, y, z\} \leq \frac{2}{3}$.

Problem 8. Square $ARML$ has sides of length 11. Collinear points X, Y, and Z are in the interior of $ARML$ such that $\triangle LXY$ and $\triangle RXZ$ are equilateral. Given that $YZ = 2$, compute $[RLX]$.

Problem 9. Compute the number of permutations $x_1, x_2, \ldots, x_{10}$ of the integers $-3, -2, -1, \ldots, 6$ that satisfy the chain of inequalities

$$x_1x_2 \leq x_2x_3 \leq \cdots \leq x_9x_{10}.$$

Problem 10. Compute the greatest integer n for which the decimal representation of $\frac{1000}{n}$ is of the form $\underline{1}.\underline{T}\,\underline{A}\,\underline{S}\,\underline{T}\,\underline{Y}\,\underline{7}\ldots$, where the digits T, A, S, and Y are not necessarily distinct.